

Birla Institute of Technology & Science, Pilani
1st Semester 2019-2020
Comprehensive Exam-Part A (Set I) (Close Book)

Course No: EEE F311
Maximum Marks: 20

Course Title: Communication Systems
Duration: 60 Minutes

Date: 09-12-2019

Please write the final answer in the space provided.
Rough work should be done in the main answer script.

- Q1. A phase modulated (PM) signal is given as $x(t) = 20 \cos(200\pi t + 12\pi m(t))$ where the message signal $m(t) = \Delta(t/6)$ and Δ denotes the triangular signal. Find the peak frequency deviation (in Hz) if the carrier frequency is 100 Hz.

Ans:

2

- Q2. There are 7 alphabets with probabilities $\{0.05, 0.1, 0.15, 0.17, 0.18, 0.22, 0.13\}$. Using a ternary Huffman code, find the average length of code.

Ans:

1.78

- Q3. A generator polynomial $g(x) = x^2 + 1$ construct a $(4, 2)$ cyclic code. Find the error detecting capability of the code.

Ans:

1

- Q4. Write down two types of single-user spread spectrum techniques.

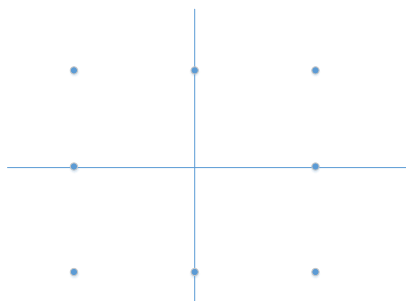
Ans:

Frequency Hopping [1 Mark] and Direct Sequence [1 Mark]

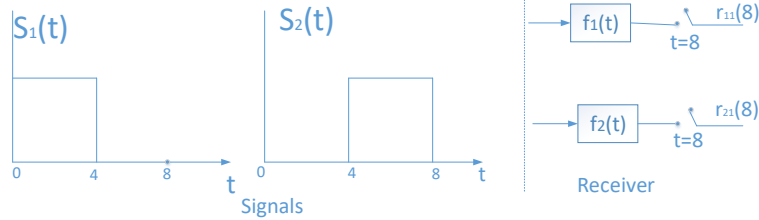
- Q5. Determine the average transmitted power for an 8-point QAM signal constellation as given in the figure below. The minimum distance between two adjacent constellation point is $d_{min} = 2$.

Ans:

Symbol power= 6 or Bit power= 2



- Q6. In a binary transmission rectangular pulses $s_1(t)$ and $s_2(t)$ are transmitted under noiseless channel condition. Receiver uses two matched filters $f_1(t)$ and $f_2(t)$ corresponding to $s_1(t)$ and $s_2(t)$. Assuming that $s_1(t)$ was transmitted, find $\frac{r_{11}(8)}{r_{21}(8)}$, where $r(8)$ denotes the received signal at $t = 8$.



Ans:

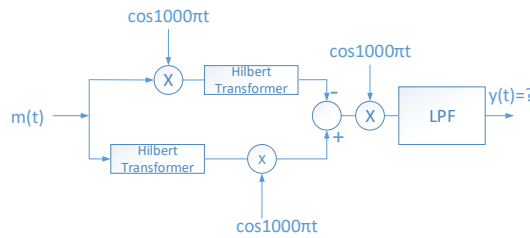
$$\infty$$

- Q7. In an AWGN channel $y(t) = x(t) + n(t)$, the signal power is fixed at 20dBm and the noise PSD $N_0/2 = -120$ dBW/Hz. Find the channel capacity (in bps) if the channel bandwidth becomes unbounded i.e., $B \rightarrow \infty$.

Ans:

$$0.72 \times 10^{11} [2 \text{ Marks}] \text{ or } 1.44 \times 10^{11} [1 \text{ Mark}]$$

- Q8. Find $y(t)$ in terms of $m(t)$ as shown in the system diagram below. The frequency response of both message signal $m(t)$ and LPF is $\Pi(f/200)$.



Ans:

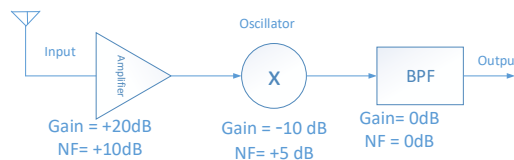
$$\frac{m_h(t)}{2}$$

- Q9. A voice signal has a frequency range 300 Hz- 3300 Hz. It is sampled at the Nyquist rate. A 4-bits/sample PCM encoding ensures transmission at the channel capacity. For data transmission, M-ary transmission using Nyquist pulses with a roll-off factor 0.5 is used. Find the value M if the channel bandwidth is 3000 Hz.

Ans:

$$64$$

- Q10. A part of super-hetrodyne receiver is shown in the figure below. The noise PSD (two sided) at the input of amplifier is $N_0/2 = -100$ dBm/Hz. Find the noise power (in dBm) at the output of band-pass filter. The bandwidth of BPF is 50 Hz.



Ans:

$$-59.9906$$

Solution: Detailed

Q1. $\Delta_f = \max \frac{1}{2\pi} \frac{d}{dt} 12\pi m(t) = 6 \max \frac{d}{dt} m(t) = 6 \times (2/6) = 2.$

Ans:

2

Q2. Combine three alphabets at one time, and coding as 012 using Huffman code gives codes as 0, 10, 11, 12, 20, 21, 22. Average length = $0.22 + 2 \times (0.18 + 0.17 + 0.15 + 0.13 + 0.10 + 0.05) = 1.78$

Ans:

1.78

Q3. Codes $c(x) = d(x)g(x)$: When $d = 00$, code = 0000, when $d = 01$, code = 0101, when $d = 10$, code = 1010, when $d = 11$, code = 1111. $d_{min} = 2$, error detection $d_{min} = t + 1$ gives $t = 1$.

Ans:

1

Q4. Write down two types of single-user spread spectrum techniques.

Ans:

Frequency Hopping and Direct Sequence

Q5. Signal points are at distance 2 and $2\sqrt{2}$ units from the origin. Average power $\frac{1}{8}[4 \times 2^2 + 4 \times (2\sqrt{2})^2] = 6$. Average bit power = $6/3=2$.

Ans:

Symbol power= 6 or Bit power= 2

Q6. Output signal $r_{21}(8) = f_2(t) * s_1(t) = 0$ at $t = 8$. Thus the ratio will be infinity.

Ans:

∞

Q7. Capacity $C = B \log_2(1 + S/N_0B)$. When limit $B \rightarrow \infty$, $C = 1.44S/N_0$. Converting in linear scales, gives the result

Ans:

$$1.44 \frac{10^2 \times 10^{-3}}{2 \times 10^{-12}} = 0.72 \times 10^{11}.$$

Q8. Writing the output signals and noting that LPF passes only the signal within the BW of $m(t)$:

Ans:

$$\frac{m_h(t)}{2}$$

Q9. Nyquist rate 6600 samples/second. PCM output= capacity= 6000 samples/second $\times$ 4 bits/sample = 26400 bps. Using BW = 3000 and $\alpha = 0.5$, $R_s = \frac{2B}{1+\alpha} = 4000$ symbols/ second. Each symbol should carry 6.6 bits. Hence $M = 2^6 = 64$. **We can not take $b = 7$ or $M = 128$ since it will exceed the channel capacity.**

Ans:

64

Q10. Resultant NF combined with amplifier and oscillator (since BPF is noiseless) $= 10 + \frac{10^{0.5}-1}{100} = 10.0216$. Output power (product of noise PSD with BW of BPF and gains in linear scale 100 (amplifier) and 10 (oscillator) and NF (10.0216)) $= 10^{-10} \times 2B \times 10.0216 \times 100 \times 1/10 = 1.0022 \times 10^{-6} mW = -59.9906 \text{ dBm}$.

Ans:

-59.9906